

eSim for Circuit Design, Simulation, Analysis and PCB Design

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In this article we take a look at eSim, a powerful open source software for integrated electronic design applications including circuit design, simulation, analysis and PCB design. The software is built with the help of popular open source software such as Ngspice, KiCad, Scilab and Python.

It is developed as an open source alternative to commercial software like OrCAD, Xpediton and HSPICE. Educational institutions and small and medium enterprises can effectively use this software that offers the user experience and capabilities equivalent to proprietary ones.

An all-in-one EDA package

There are a series of steps involved in electronic design automation (EDA). In the first step, specifications are laid down. These specifications take the shape of a design that could be in various forms such as a schematic circuit or a logical description in a High-level Description Language (HDL). The design is then simulated and redesigned, if necessary.

Once specifications are met, the design is converted into a PCB layout, a chip layout or ported to a field programmable gate array (FPGA). The product

thus obtained is again checked to see if it meets the specifications and, for that, the entire cycle is repeated till desired results are obtained.

A large number of proprietary EDA tools including Cadence, Synopsys, Mentor Graphics and Xilinx can perform all these jobs in a comprehensive fashion. Unfortunately, the cost associated with the purchase of

such proprietary design tools discourages the student community from colleges as well as SMEs from playing around with such tools.

eSim, previously known as Oscad or FreeEDA, took its shape in IIT Bombay under free and open source software in science and engineering education (FOSSEE) group, making the software a viable alternative to open source ones.

Advanced features

Rakhi R. and Kannan M. Moudgalya in their article titled, 'Oscad: Open Source Computer Aided Design Tool' discuss the advanced features of this tool. Let us take a quick glance at these.

Model Builder. How well the devices in the simulations are modelled is a prime determinant factor of the accuracy of the simulation. A set of device models including bipolar junction diodes, bipolar junction transistors (BJTs) and field effect transistors (FETs) are provided.

Suppose we are dealing with a bridge rectifier circuit. Parameters in the device model (say, the diode) can be set using the model builder tool and can even be reused by exporting the design into other files.

Subcircuit Builder. A complex circuit can be broken into simpler sub-circuits and could be reused in other projects. This ensures scalability and modularity of the design.

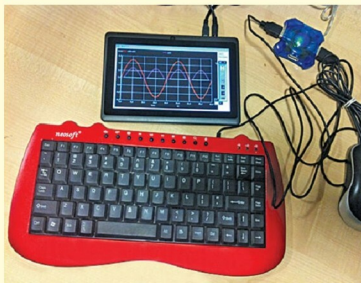


Fig. 1: eSim software running on Aakash tablet (Image courtesy: www.aakashlabs.org/gnu)

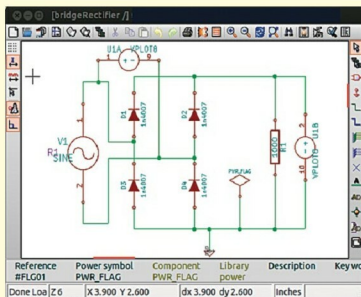


Fig. 2: Simulating a rectifier circuit using eSim